Enterprise-Grade Full Stack Application

A Project Report submitted in partial fulfilment of the requirements for the award of the degree of

Master of Computer Applications

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Declaration

I hereby declare that the work which is being presented in the MCA Project "Full Stack Development Internship at Manthan IT Solutions", in partial fulfillment of the requirements for the award of the Master of Computer Applications and submitted to the Department of Computer Engineering and Applications of GLA University, Mathura, is an authentic record of my own work carried under the supervision of Sunanda Biswas.

The contents of this project report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree.

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CERTIFICATE

This is to certify that the above statements made by the candidates are correct to the best of my knowledge and belief.

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I would like to express my profound gratitude to everyone who contributed to the successful completion of my six-month internship at Manthan IT Solutions. I am deeply thankful to my mentors, technical leads, and the entire engineering team for their exceptional guidance, unwavering support, and the invaluable knowledge they imparted to me throughout this enriching training period. Their profound insights into full-stack software engineering, GoLang backend architecture, React-based frontend development, and agile project management have been instrumental in shaping my professional learning curve and instilling in me a disciplined approach to software development. Furthermore, I extend my sincere appreciation to the management and human resources department at Manthan IT Solutions for providing me with this remarkable opportunity to contribute to real-world enterprise products spanning corporate reporting, travel booking, and telephony management domains. Finally, I would like to acknowledge the academic faculty at my university, whose foundational teachings provided the theoretical framework that made this practical application possible.

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ABSTRACT

This comprehensive technical report documents my six-month professional experience as a Full Stack Developer at Manthan IT Solutions. The internship served as an immersive, practical platform designed to bridge the theoretical concepts acquired academically with complex, industry-standard full-stack software engineering practices. Over the course of this tenure, I made substantial, independent contributions to three primary enterprise-grade applications: the Project Work Opt-In/Opt-Out Report Website, the Travel Website, and the Calling Website. To address the diverse and highly complex computational requirements of these projects, my technical stack encompassed GoLang for high-concurrency backend services, React.js for interactive and component-driven frontend interfaces, MySQL for normalized relational data modelling, Lucid for architectural and workflow visualization, and Framer Motion for production-grade UI animations. This document provides a detailed exposition of the system architectures deployed, the normalized MySQL database schemas engineered, the RESTful API design principles strictly adhered to, and the Role-Based Access Control (RBAC) authorization models implemented in GoLang. Additionally, this report elucidates the integration of advanced mechanisms including goroutine-based concurrency for real-time telephony management, WebSocket-driven live agent dashboards, and finite-state machine modelling for the complete call lifecycle. By thoroughly detailing the Software Development Life Cycle (SDLC) adhered to across the corporate, travel, and telephony domains—including continuous integration principles and rigorous quality assurance testing—this report demonstrates a consolidated foundation in scalable full-stack application development, high-concurrency backend architecture, and polished frontend engineering.

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Chapter 1

Introduction

1.1 Motivation and Overview

The six-month professional engagement at Manthan IT Solutions served as a pivotal transition from the academic study of software principles to the rigorous discipline of full-stack product engineering. Operating within a fast-paced, Agile-driven delivery environment, my designated role was that of a Full Stack Developer. The primary motivation behind undertaking this position was to acquire deep, hands-on experience in designing performant backend services using GoLang, building modern interactive user interfaces with React, modelling normalized relational data in MySQL, architecting application flow with Lucid diagrams, and crafting fluid, production-grade animations using Framer Motion. My responsibilities encompassed the end-to-end design, development, optimization, and deployment of secure and highly scalable web applications spanning corporate reporting, travel booking, and telephony management domains.

1.2 Objective Of The Report

The principal objective of this technical report is to present a detailed, cohesive overview of the engineering responsibilities, architectural decisions, and software projects executed during my tenure as a Full Stack Developer at Manthan IT Solutions. Specifically, this report intends to:

Formally document the technical methodologies, design patterns, and full-stack engineering practices acquired across the six-month duration.

Provide a granular, component-level analysis of the end-to-end architecture, normalized MySQL schemas, and core algorithmic feature sets of the three primary enterprise-grade applications: the Project Work Opt-In/Opt-Out Report Website, the Travel Website, and the Calling Website.

Elucidate the engineering paradigms and frameworks applied to resolve real-world challenges such as concurrent request handling in Go, state synchronization in React, fluid UI motion design with Framer Motion, and visual system modelling with Lucid.

Fulfill the academic requirements of the curriculum by demonstrating professional competency in the complete lifecycle of full-stack software development.

1.3 Issues And Challenges In Modern Full Stack Development

Modern full-stack development confronts unprecedented challenges related to concurrency, real-time interactivity, security, and cross-stack consistency. Throughout the engagement, key challenges included designing Go services capable of handling thousands of concurrent goroutines without race conditions, preventing the well-known N+1 query problem in MySQL through eager loading and indexed joins, harmonizing optimistic UI updates in React with eventual server consistency, ensuring uniform animation performance with Framer Motion across low-end devices, and enforcing stateless authentication using cryptographically signed JSON Web Tokens. Addressing these challenges required a paradigm shift from procedural request-response thinking to event-driven, component-oriented, and concurrency-safe engineering.

Chapter 2

Software Development

2.1 Agile Methodology and The Scrum Framework

The entire software development lifecycle across all assigned projects at Manthan IT Solutions was strictly governed by the Agile methodology, employing the Scrum framework. This iterative approach facilitated continuous integration, regular stakeholder feedback, and rapid course-correction in response to evolving client requirements.

Sprint Lifecycle: Development was compartmentalized into rigid two-week sprints. Each cycle began with a Sprint Planning meeting wherein product backlog items were analyzed, technically estimated using story points, and assigned to specific engineers across the frontend and backend layers.

Daily Synchronization: A concise, time-boxed daily stand-up was conducted to report progress, outline daily engineering objectives, and identify technical blockers requiring architectural discussion or pair-programming intervention.

Sprint Review and Retrospective: Each sprint concluded with a demonstration of newly completed React features and Go API endpoints to stakeholders in a staging environment, followed by an internal retrospective to refine workflow, address technical debt, and improve CI/CD pipelines for subsequent iterations.

2.2 Advanced Version Control System and Collaboration

Git was rigorously utilized for source code management to ensure seamless, conflict-free collaboration across the distributed engineering team at Manthan IT Solutions.

Branching Strategy: The team adhered to the Git Flow methodology. The master branch exclusively housed production-ready builds. The develop branch served as the primary integration hub, while individual engineers worked exclusively within isolated feature or bugfix branches to prevent contamination of the primary codebase.

Peer Review and Pull Requests: Source code was merged into develop only after exhaustive peer code reviews via Pull Requests. This process enforced high code quality, identified potential goroutine leaks in Go services, flagged unnecessary re-renders in React components, and ensured strict adherence to organizational styling and linting guidelines.

2.3 Issue Tracking and Technical Documentation

Platform tools such as Jira and Trello were extensively utilized for the agile tracking of bugs, granular engineering tasks, and user stories. Every distinct API endpoint, MySQL migration, React component, or Framer Motion animation requirement was documented and tracked as an actionable ticket. Lucid was used for maintaining visual architectural documentation, including entity-relationship diagrams, data-flow diagrams, and component hierarchy charts, allowing the team to maintain a single source of truth for system design.

Chapter 3

System Requirements Analysis

Prior to the commencement of any feature development, a rigorous and exhaustive analysis of the system requirements was executed for each project to guarantee that the underlying full-stack infrastructure could autonomously support the projected user load, data throughput, and interactive richness demanded by the client.

3.1 Functional Requirements

Functional requirements delineated the precise operational capabilities and business logic the system must inherently possess. Across the three projects, core functional requirements included:

Highly secure user authentication and registration utilizing bcrypt for password salting and hashing, combined with JWT-based session management implemented in GoLang.

The dynamic computation and rendering of role-specific dashboards in React, enforcing strict data visibility based on organizational hierarchy and opt-in/opt-out preferences.

Processing of complex multipart form submissions containing user documents, travel itineraries, and call recordings, with server-side validation in Go.

The capability to systematically export and import large MySQL datasets via Excel (XLSX) and CSV parsers for reporting and bulk data migration.

Automated, asynchronous SMTP-based email notifications and in-app real-time alerts using WebSockets, triggered upon specific state changes in the database.

Smooth, performance-optimized page transitions and micro-interactions powered by Framer Motion, providing a polished, premium feel across all client-facing interfaces.

3.2 Non-Functional Requirements

Security Architecture: Comprehensive protection mechanisms were implemented against SQL Injection (via parameterized queries in Go's database/sql package), Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF). API rate-limiting, using token-bucket algorithms in Go middleware, was enforced to mitigate Distributed Denial of Service (DDoS) attempts and brute-force login attacks.

Scalability and Concurrency: The Go-based backend was mandated to seamlessly support thousands of concurrent HTTP requests by leveraging goroutines and channels. This was particularly critical for the Calling Website, where hundreds of agents could initiate or receive calls simultaneously.

Performance Thresholds: A strict engineering baseline was established requiring all MySQL queries to execute and return within two hundred milliseconds. This necessitated B-Tree indexing on foreign keys and high-traffic columns, alongside selective denormalization of reporting tables to avoid expensive multi-table joins.

Reliability and High Availability: The infrastructure demanded ninety-nine point nine percent (99.9%) uptime. This was achieved through robust global error handling in Go, circuit breaker patterns for outbound calls, and graceful fallback UI states in React, ensuring isolated backend exceptions did not precipitate full-system failures.

Chapter 4

Technologies Utilized

4.1 GoLang and Concurrent Backend Execution

GoLang was selected as the primary backend programming language across all three projects due to its statically typed compilation, exceptional runtime performance, and first-class support for concurrency through goroutines and channels. Unlike traditional thread-based runtimes, Go's lightweight goroutines permitted the spawning of tens of thousands of concurrent execution contexts with minimal memory overhead. The standard net/http package, combined with lightweight routing libraries, provided a performant and idiomatic foundation for constructing RESTful APIs. Go modules were utilized for deterministic dependency management, and structured logging via zap or zerolog ensured production-grade observability.

4.2 React.js and the Virtual DOM

React.js served as the foundational frontend library across all client-facing interfaces. Its component-based architecture, unidirectional data flow, and reconciliation algorithm for the Virtual DOM enabled the construction of highly interactive and maintainable user interfaces. React Hooks such as useState, useEffect, useMemo, and useCallback were utilized extensively to manage local and derived state, while Context API and Redux Toolkit handled cross-cutting global state. The asynchronous data-fetching layer was integrated with the Go backend through Axios, with normalized JSON contracts agreed upon between the frontend and backend teams.

4.3 MySQL and Relational Data Modelling

MySQL constituted the foundational relational database management system powering all persistent data operations. Database schemas were meticulously designed up to the Third Normal

Form (3NF) to eradicate redundancy, protect data integrity, and prevent insert, update, and delete anomalies. ACID compliance was strictly guaranteed, with multi-table operations such as user creation, role assignment, and initial workspace generation executed within single transactional blocks utilizing rollback mechanisms in case of partial failure. Strategic B-Tree indexing on foreign keys and frequently filtered columns ensured query latencies remained well within agreed performance thresholds.

4.4 Lucid for Architectural Visualization

Lucid (Lucidchart) was extensively utilized as the visual modelling and architectural diagramming tool across the engineering lifecycle. Prior to writing any production code, system architecture diagrams, database entity-relationship diagrams, sequence diagrams, and user-journey flowcharts were drafted in Lucid and shared with both technical and non-technical stakeholders. This visual-first approach drastically reduced ambiguity in requirement translation, exposed design flaws early, and produced living documentation that could be updated alongside code changes. Lucid's collaborative editing features further allowed engineers, designers, and product managers to iterate on architecture in real time.

4.5 Framer Motion for Production-Grade Animation

Framer Motion was selected as the dedicated animation library for the React frontend across all three projects. It provided a declarative, physics-aware API for choreographing component entry and exit transitions, page-level route animations, drag-and-drop interactions, and complex shared layout transitions. By leveraging Framer Motion's variants, AnimatePresence, and layout animations, the user interface achieved a fluid, native-application feel while preserving accessibility and respecting reduced-motion preferences. Performance was carefully tuned by limiting animations to GPU-accelerated transform and opacity properties, preventing layout thrashing on lower-end devices.

Chapter 5

Project 1 - Project Work Opt-In/Opt-Out Report Website

5.1 Introduction And Objective

The Project Work Opt-In/Opt-Out Report Website was the inaugural project of my engagement at Manthan IT Solutions. It was engineered as a comprehensive internal reporting platform designed to allow employees to formally opt in or opt out of upcoming project allocations, while simultaneously providing managers and HR personnel with rich analytical dashboards summarizing participation, capacity, and skill alignment across the organization. The system was architected to serve as the single source of truth for project staffing decisions.

5.2 System Architecture And Technical Stack

The system utilized a clean, layered REST API design. The React client communicated via secure HTTPS to the GoLang application layer. The backend enforced strict request validation through custom Go struct tags and validator libraries, and authorization via JWT cryptographic tokens. Upon successful validation, the service layer executed parameterized SQL queries against the MySQL database, thereby natively preventing SQL injection. Architectural diagrams describing the request lifecycle, the opt-in workflow, and the reporting pipeline were authored and continuously maintained in Lucid.

5.3 Database Schema Design And Entity Relationships

The relational schema was highly structured and interconnected. Core MySQL tables included:

Users Table: Tracking primary identification, hashed credentials, departmental associations, and skill metadata.

Projects Table: Storing project metadata, required skills, capacity, and scheduled timelines.

Opt_Status Table: Maintaining a state-machine record of each user's opt-in or opt-out decision per project, alongside justification text and timestamps.

Reports Table: Materialized aggregations used by management dashboards, refreshed on a scheduled cadence to keep dashboard latency low.

5.4 Key Modules And Algorithmic Workflow

Opt-In/Opt-Out Workflow: When an employee submitted an opt-in or opt-out decision, the Go backend validated the user's eligibility, ensured that the deadline window was still open, persisted the decision atomically in MySQL, and dispatched an asynchronous notification to the project manager via a background goroutine.

Reporting Engine: A reporting service aggregated opt-in counts, capacity gaps, and skill coverage per project. The aggregation was implemented using optimized GROUP BY queries with covering indexes in MySQL, and the resulting datasets were rendered in the React dashboard with smooth transitions powered by Framer Motion to convey changes between filter selections.

Bulk Data Import: To onboard organizational data efficiently, an Excel upload endpoint was implemented in Go using stream-based parsing. Each row was validated, normalized, and inserted via batched, transactional INSERT statements, processing thousands of records within seconds.

Chapter 6

Project 2 - Travel Website

6.1 Introduction And Scope

The Travel Website is a customer-facing platform engineered at Manthan IT Solutions to enable end users to discover, plan, and book travel packages spanning flights, hotels, and curated multi-city itineraries. The platform was designed to deliver a visually rich, animation-driven browsing experience while remaining fast, accessible, and resilient under heavy concurrent load.

6.2 Architecture And Frontend Experience

The backend ecosystem was constructed in GoLang, exposing a well-versioned REST API consumed by a React single-page application. MySQL served as the primary persistence layer for users, destinations, packages, and bookings. On the frontend, Framer Motion was leveraged extensively to deliver cinematic page transitions between the home page, destination detail pages, and the booking funnel. Image galleries used shared layout animations so that destination thumbnails smoothly morphed into full-screen hero presentations, significantly improving perceived performance and user engagement.

6.3 Search, Filtering, and Booking Workflow

A high-performance search service was implemented in Go, exposing endpoints for full-text search over destinations, faceted filtering on price range, duration, rating, and category, and pagination via cursor-based keyset pagination to maintain sub-200ms response times even on large datasets. The booking workflow was designed as a strict multi-step state machine: package selection, traveller information, payment, and confirmation. Each transition was validated server-side in Go,

persisted within a single MySQL transaction, and reflected on the React UI through animated step indicators built with Framer Motion.

6.4 Database Schema and Data Integrity

The MySQL schema for the Travel Website included tables for destinations, packages, package_inclusions, bookings, payments, and traveller_details. Foreign key constraints were strictly enforced to guarantee referential integrity. Sensitive payment metadata was tokenized and stored separately, with the actual transaction handled via a third-party payment gateway whose webhooks were verified through HMAC signatures inside the Go service, ensuring that booking confirmation could not be forged.

6.5 Visual System Design with Lucid

Before development began, the complete user journey, from landing page to post-booking confirmation email, was mapped in Lucid as a sequence diagram and a state diagram. Database entity-relationship diagrams were also maintained in Lucid and updated on every schema migration, ensuring that newly onboarded engineers could understand the data model within minutes.

Chapter 7

Project 3 - Calling Website

7.1 Introduction And Objective

The Calling Website is an internal telephony and call-management platform engineered for an enterprise client of Manthan IT Solutions. The platform consolidates inbound and outbound calling, agent assignment, call recording metadata, follow-up scheduling, and post-call analytics into a single responsive web application, replacing several disjointed legacy tools.

7.2 Real-Time Architecture and Concurrency Model

GoLang's native concurrency primitives were the cornerstone of this project. Each active call session was represented as a lightweight goroutine that managed its own state machine: initiating, ringing, connected, on-hold, and terminated. Inter-goroutine communication was performed via typed channels, while a centralized supervisor ensured graceful cleanup on disconnects. The frontend, built in React, maintained a persistent WebSocket connection to the Go backend so that agents received call events, queue updates, and supervisor messages in real time without polling.

7.3 Agent Dashboard and Call Lifecycle Management

The agent dashboard, implemented in React with Framer Motion, presented a live queue of incoming calls with subtle, attention-directing animations on newly arriving items. Agents could accept, transfer, or escalate calls through clearly animated controls. Each lifecycle event was sent over the WebSocket channel to the Go backend, persisted in MySQL with millisecond-precision timestamps, and reflected back to supervisors in their monitoring view. Post-call dispositions, notes, and follow-up tasks were captured in a structured form, enabling rich downstream analytics.

7.4 Database Schema and Reporting

Core MySQL tables included agents, customers, calls, call_events, dispositions, and follow_ups. The calls table maintained denormalized summary fields such as total_duration_seconds and final_disposition to accelerate reporting queries, while the call_events table preserved a full event log for audit purposes. Indexed columns on agent_id, customer_id, and started_at allowed sub-second filtering across millions of historical call records.

7.5 Workflow Modelling in Lucid

The complete call lifecycle, including transfer, hold, conference, and wrap-up paths, was rigorously modelled as a finite-state machine in Lucid. This visual specification became the authoritative reference for both Go backend developers implementing the supervisor goroutine and React developers building the agent UI, ensuring perfect alignment of state semantics across the stack.

Chapter 8

System Testing and Quality Assurance

Rigorous quality assurance protocols were heavily integrated into the full-stack development lifecycle to ensure system stability, prevent data corruption, and guarantee uptime prior to production deployment.

8.1 API Testing and Automation

Postman was utilized extensively as the primary tool for API testing and contract verification of the GoLang services. Specialized Postman Collections were authored, with environment variables used to switch instantly between Local, Staging, and Production servers. Pre-request scripts automatically acquired and injected JWT tokens, while assertion scripts validated the exact data types and structures of JSON response payloads. In addition, Go's built-in testing package was used to author unit and integration tests covering business logic and database access layers, with table-driven tests ensuring high branch coverage.

8.2 Frontend Testing and Visual Regression

On the React side, component-level tests were authored using React Testing Library and Jest, focusing on user-observable behaviour rather than implementation detail. Critical Framer Motion animations were tested by asserting on intermediate and final accessible states, while visual regression checks were performed manually across breakpoints to ensure consistent UI fidelity on desktop, tablet, and mobile devices.

8.3 Validation, Sanitization, And Graceful Degradation

Graceful Degradation Architecture: Comprehensive global exception handlers and middleware were implemented in Go. In the event that an external dependency such as the SMS gateway,

payment processor, or SMTP server became unavailable, the backend cleanly intercepted the timeout, logged the error with full context, and delivered a structured HTTP 503 response to the React client. The frontend, in turn, displayed user-friendly fallback states, animated by Framer Motion, instead of crashing or appearing frozen.

Input Sanitization and Cyber Security: Dedicated validation layers were applied across all incoming HTTP request payloads. These layers stripped potentially malicious HTML, JavaScript, or SQL injection vectors from text fields, ensuring the MySQL database remained pristine and the application remained resilient against standard web vulnerabilities.

Chapter 9

Conclusion and Future Scope

9.1 Summary Of Professional Learnings

The transition from a theoretical academic setting to the rigorous, high-stakes professional environment at Manthan IT Solutions over the past six months has been profoundly transformative. The internship catalysed a rapid evolution in technical proficiency, analytical problem-solving, and system-level architectural thinking. Primary professional advancements include the practical mastery of GoLang for high-concurrency backends, the construction of polished React user interfaces, the design and tuning of normalized MySQL schemas, the visual specification of complex workflows in Lucid, and the orchestration of expressive yet performant interactions using Framer Motion.

9.2 Professional Growth And Accountability

Beyond pure software engineering mechanics, this internship cultivated essential professional skills. The experience of collaborating within an Agile team, translating ambiguous client directives into concrete technical specifications, and independently owning critical modules across the Opt-In/Opt-Out Report Website, the Travel Website, and the Calling Website instilled a strong sense of engineering accountability, attention to detail, and professional maturity.

9.3 Future Scope And Technological Trajectory

The technical foundation established during this tenure is robust and highly scalable. Future engineering objectives include deepening expertise in DevOps practices, specifically Docker containerization and Kubernetes orchestration for Go services. I aim to engineer fully automated Continuous Integration and Continuous Deployment pipelines using GitHub Actions.

Additionally, I intend to evolve from monolithic API designs toward distributed microservice architectures, leveraging message brokers such as NATS or Kafka to manage exponentially larger asynchronous data throughput. The challenges confronted, analyzed, and successfully resolved during this six-month engagement at Manthan IT Solutions have definitively prepared me for a dynamic and impactful career as a Full Stack Developer.